

PyMuPDF Beginner's Guide

A simple, readable guide to understanding and working with PyMuPDF

1. What is PyMuPDF?

PyMuPDF is a fast, high-performance Python library used to inspect, extract, edit, and convert PDF documents.

It is human-readable, simple, and works seamlessly with modern Python applications.

2. A Simple PyMuPDF Example

Here is what working with PyMuPDF looks like in real life:

```
import pymupdf

# Load document and select page
doc = pymupdf.open("sample.pdf")
page = doc[0] # First page (0-indexed)

# Extract plain text
text = page.get_text()

# Save page as a PNG image
pix = page.get_pixmap(dpi=150)
pix.save("page_0.png")

doc.close()
```

3. Saving Images to a Specific Folder

You can save images to a specific folder in two ways:

Option A: Direct Path (Pass folder path directly into `pix.save()`)

```
import pymupdf

pdf = pymupdf.open("Input_pdfs/002_redacted.pdf")
page = pdf[0]

pix = page.get_pixmap()
# Pass the direct target path inside pix.save()
pix.save("your_folder_name/page_0.png")
```

Option B: Safe Approach (Ensure folder exists first)

```
import os
import pymupdf

output_dir = "your_folder_name"
os.makedirs(output_dir, exist_ok=True) # Prevents FileNotFoundError if folder doesn't exist

pdf = pymupdf.open("Input_pdfs/002_redacted.pdf")
page = pdf[0]

pix = page.get_pixmap()
pix.save(os.path.join(output_dir, "page_0.png"))
```

4. What Types of Data Can You Extract?

PyMuPDF supports multiple simple extraction formats:

- **Text (String):** Plain text wrapped in standard lines.
- **Blocks:** Structured text blocks with bounding box coordinates `(x0, y0, x1, y1, text, block_no, block_type)`.
- **Words:** Individual words with location tuples `(x0, y0, x1, y1, word, block_no, line_no, word_no)`.
- **JSON / HTML:** Structured layout data containing font names, sizes, colors, and positioning.
- **Images:** Native raster graphics stored directly in PDF pages.

5. Golden Rules of PyMuPDF Usage

Remember these 3 rules to avoid syntax errors and deprecation warnings:

- Always use `import pymupdf` around your code instead of `import fitz`. Old module aliases will throw warnings in newer releases.
- Remember that page numbers are 0-based! `doc[0]` refers to the first page.
- Always call `doc.close()` or use context managers to free system memory and unblock file locks.

6. How to Get (Index) Values

When you read a PDF in Python code, you get values by calling page extraction methods.

PYTHON EXTRACTION WORKFLOW

```
import pymupdf

doc = pymupdf.open("document.pdf")
page = doc[0]

# 1. Get structured text blocks
blocks = page.get_text("blocks")
print(blocks[0][4]) # Output: Text of first block

# 2. Get individual word coordinates
words = page.get_text("words")

# 3. Render page to image bytes
pix = page.get_pixmap(dpi=150)
img_bytes = pix.tobytes("png")
```

7. How to Change or Redact a Specific Value

To update or permanently remove text, target its location coordinates and apply redactions.

PYTHON UPDATE WORKFLOW

```
import pymupdf

# Step 1: Parse the PDF file
doc = pymupdf.open("input.pdf")
page = doc[0]

# Step 2: Search for target text coordinates
matches = page.search_for("CONFIDENTIAL")

# Step 3: Mark target boxes for redaction
for rect in matches:
    page.add_redact_annot(rect, fill=(0, 0, 0))

# Step 4: Apply permanent redaction and save
page.apply_redactions()
doc.save("updated.pdf")
```

8. Modern API Cheat Sheet

Deprecated / Legacy	Modern Replacement
<code>import-fitz</code>	<code>import pymupdf</code>
<code>doc.pageCount</code>	<code>doc.page_count</code>
<code>doc.newPage()</code>	<code>doc.new_page()</code>
<code>pix.getPNGdata()</code>	<code>pix.tobytes("png")</code>
<code>page.getText()</code>	<code>page.get_text()</code>